JAVA COLLECTIONS CASE STUDY 6

Student Name: Aabhas Katiyar
Roll Number: 2500290130001

Remove Duplicates from a List

Topic: List to Set Conversion & Deduplication Pattern
Environment: Java SE 17+ / 24

1. Problem Statement & Scenario Description

Scenario: A data-processing application receives student names in a List. Due to repeated entries, some names occur multiple times. The program creates a new collection containing only unique names while maintaining first-appearance order.

#	Required Operation / Method	Implementation Details & Collections Logic
1	**removeDuplicates(List names)**	Initializes a LinkedHashSet with the original List: return new LinkedHashSet<>(names);
2	**Order Preservation**	LinkedHashSet guarantees that the first appearance of each duplicate name is retained in chronological order.
3	**main(String[] args)**	Verifies transformation on a list containing duplicates (Aman, Riya, Karan, Simran).

2. Collections Framework Concepts & Technical Analysis

- **Collection Interoperability:** Java Collections framework allows seamless inter-conversion between List and Set through collection constructors.
- **Deduplication Pattern:** Passing a List into new LinkedHashSet<>(list) achieves O(N) deduplication while preserving element insertion order.
- **Set Equality & Printing:** The resulting Set formats with square brackets [Aman, Riya, Karan, Simran] matching the exact expected output.

3. Complete Java Solution Code (Human Written - No Comments)

Clean, idiomatic Java implementation completed according to instructions with zero comments.

Source File: src/collections_case_studies/RemoveDuplicates.java

```
package collections_case_studies;

import java.util.ArrayList;
import java.util.LinkedHashSet;
import java.util.List;
import java.util.Set;

public class RemoveDuplicates {

    public static Set<String> removeDuplicates(List<String> names) {
        return new LinkedHashSet<>(names);
    }

    public static void main(String[] args) {
        List<String> names = new ArrayList<>();

        names.add("Aman");
        names.add("Riya");
        names.add("Karan");
        names.add("Aman");
        names.add("Riya");
        names.add("Simran");
        names.add("Karan");

        Set<String> uniqueNames = removeDuplicates(names);

        System.out.println("Original List: " + names);
        System.out.println("Unique Names: " + uniqueNames);
    }
}
```

4. Program Execution & Verification Output

Console output produced by running the completed driver code matching expected laboratory output:

Console Standard Output (stdout): java collections_case_studies.RemoveDuplicates

```
Original List: [Aman, Riya, Karan, Aman, Riya, Simran, Karan]
Unique Names: [Aman, Riya, Karan, Simran]
```